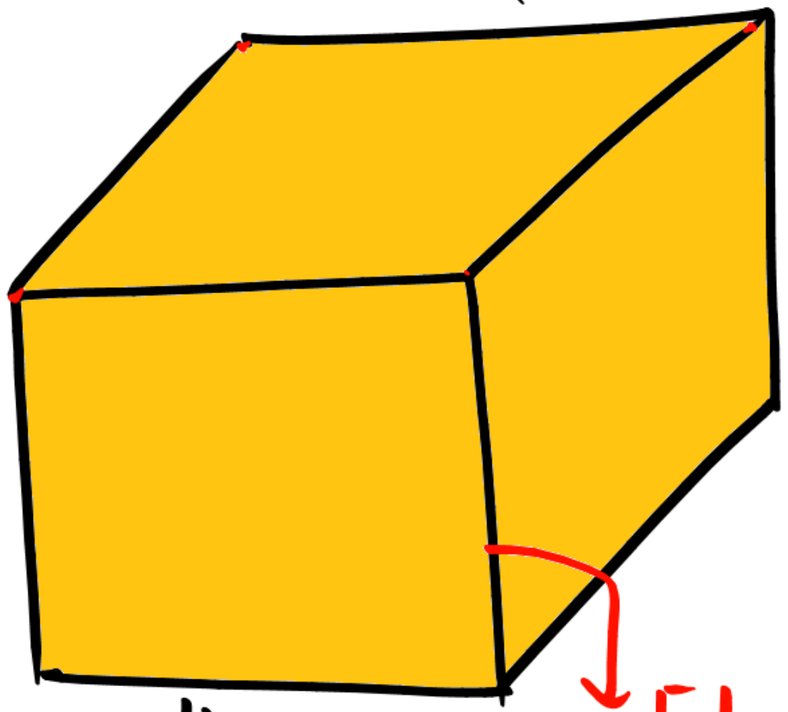
Cube and Cuboid

(घन और घनाभ)

① Surface (सतह) = 6

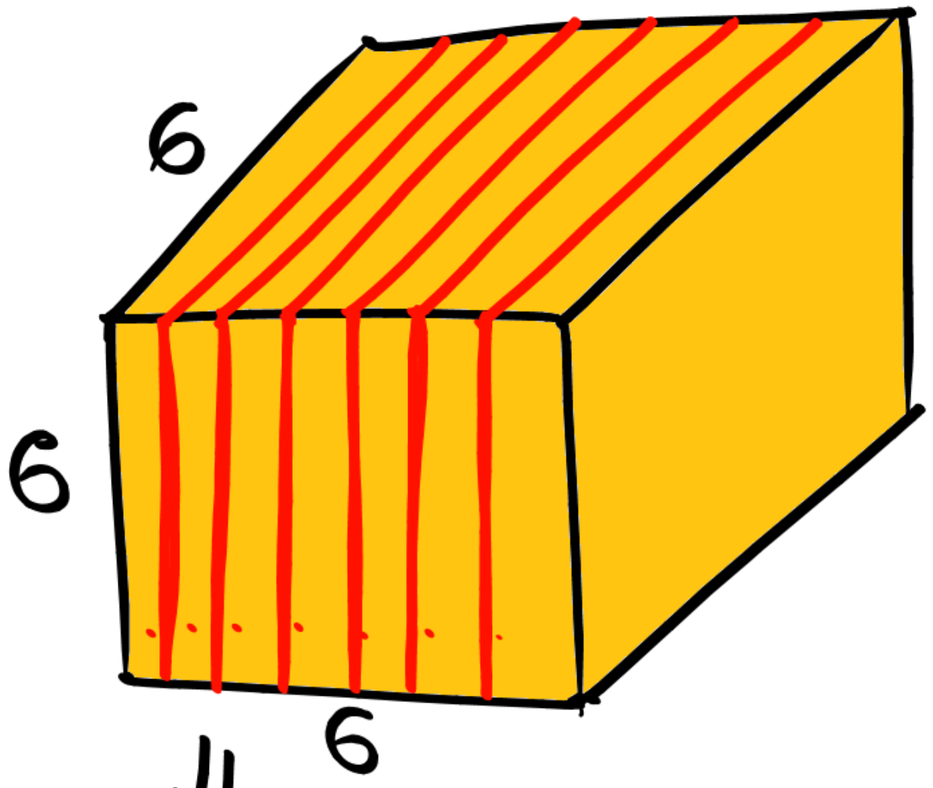
② Edges (किनारें) = 12
(Sides)

③ Corners (शीर्ष) = 8



⇓

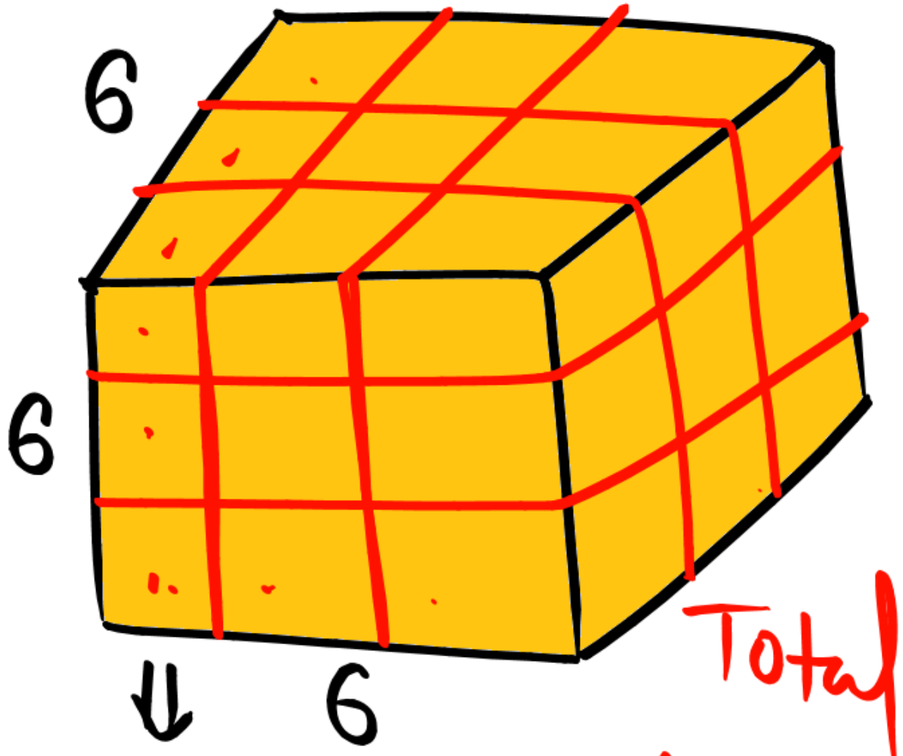
Smaller Cubes
(छोटे-2 घन)



minimum Parts Total = 7

Cuts = 6

Parts(n) = 7



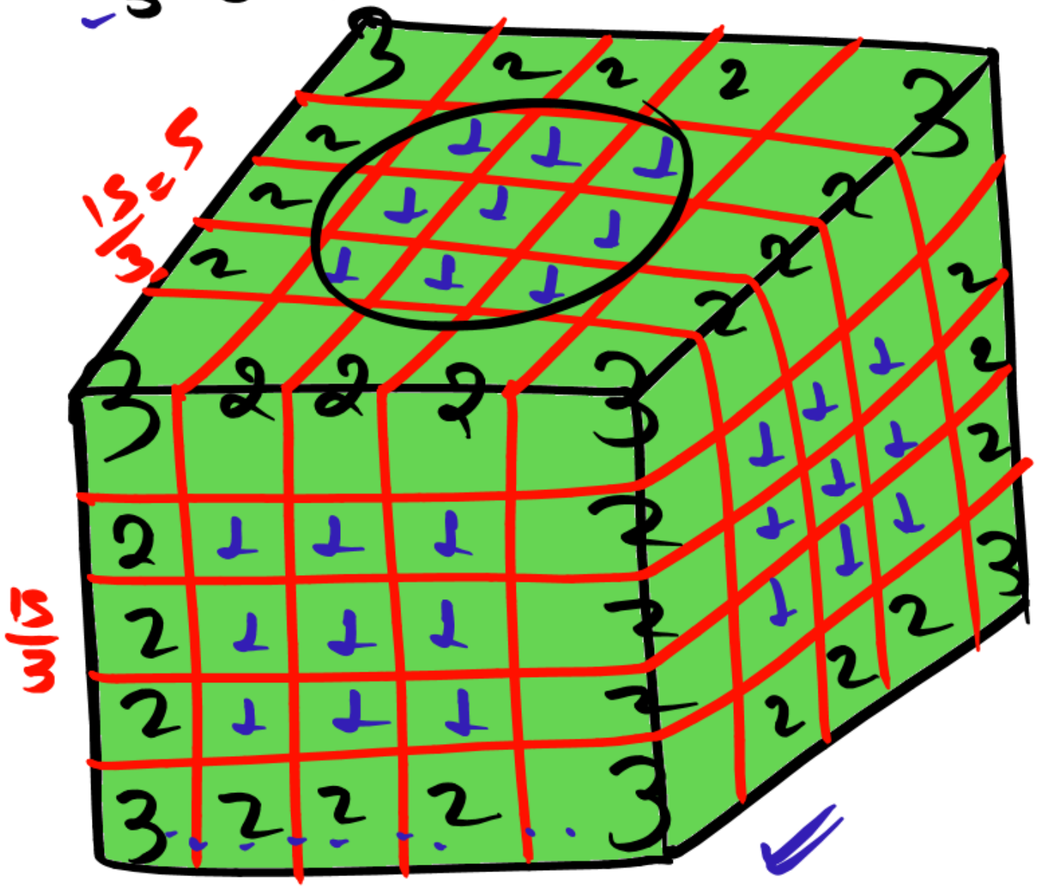
Total = 3^3 = 27

Maximum Parts (Smaller Cubes)

Cuts = 6

Parts(n) = 3,3,3

$\frac{15}{3} \times \frac{15}{3} \times \frac{15}{3} = 5$

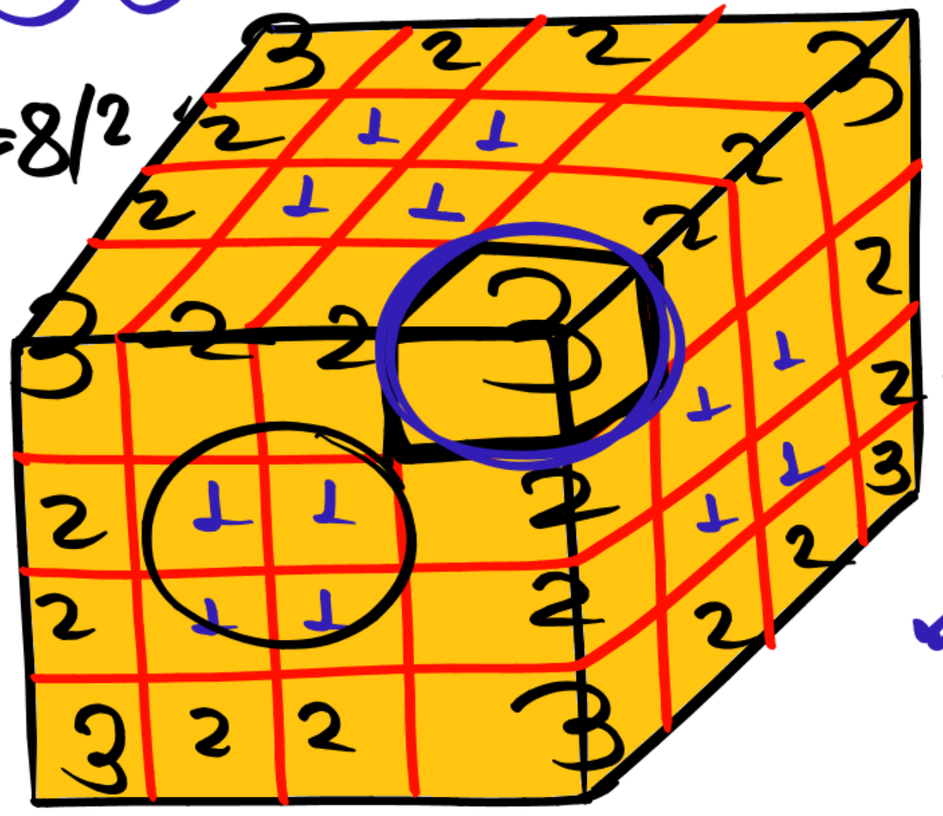


$\frac{15}{3} = 5$

$\frac{15}{3} = 5$

$4 = \frac{8}{2}$

$4 = 8/2$



$\frac{8}{2} = 4$

Types of Smaller Cubes

① Corner Cubes (शीर्ष घन) [3] = 8 ✓

② Middle/Edge Cubes (मध्य घन) [2] = $12(n-2)$ ✓

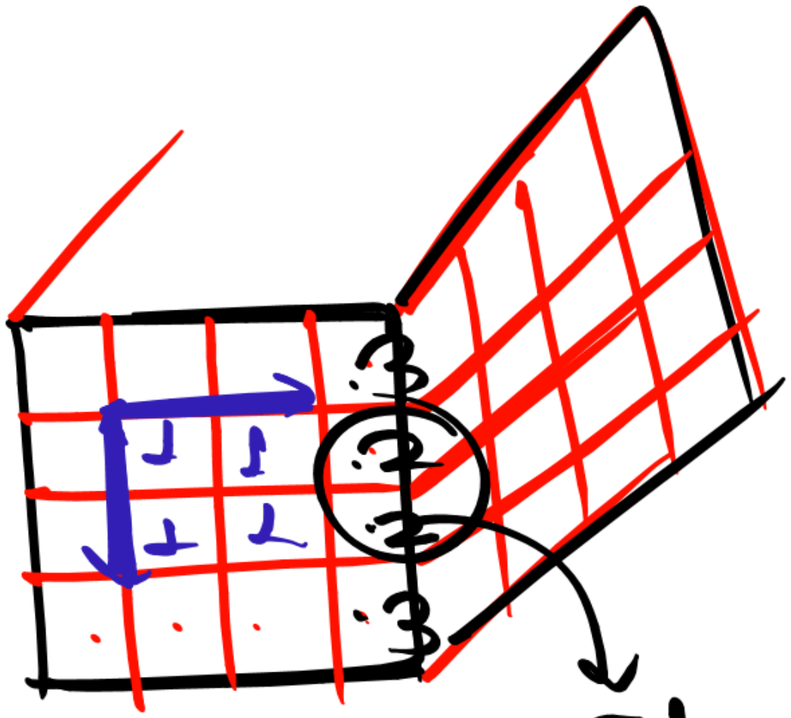
③ Central Cubes (केन्द्रीय घन) [1] = $6(n-2)^2$ ✓

④ Inner-Central/Colourless (रंगहीन) [0] = $(n-2)^3$ ✓

Total (कुल) = n^3 ✓

Total Cuts = $3(n-1)$ ✓

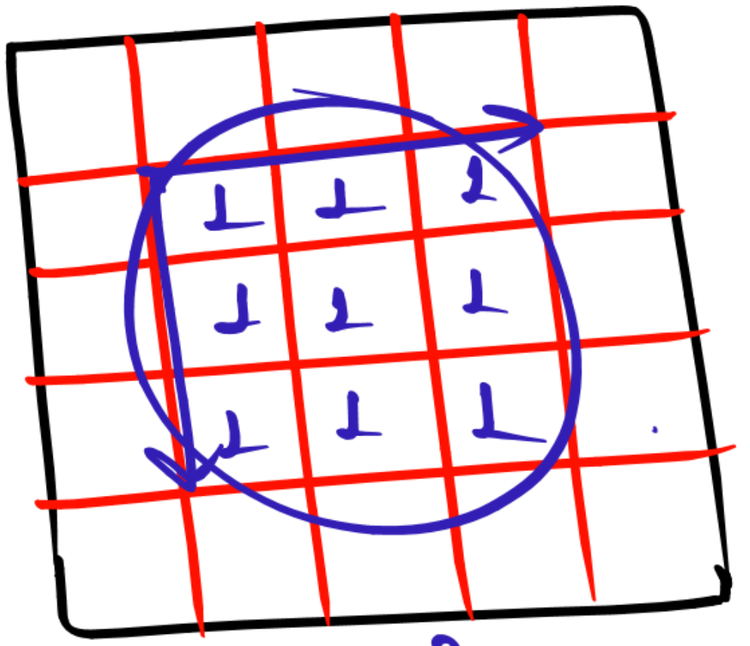
$$n = \frac{\text{big Cube length}}{\text{Small Cube length}}$$



Edge

$2 = (n-2)$

$n=5$



$(n-2)^2$

- 3 = 8
- 2 = $12(n-2)$
- 1 = $6(n-2)^2$
- 0 = $(n-2)^3$
- Total = n^3
- Cut 3 = $3(n-1)$
- $n = \frac{l_1}{l_2}$ $\frac{\text{बड़े चक्र की भुजा}}{\text{छोटे चक्र की भुजा}}$